

Automatic Grey Water Lifting Unit

AmaDrainer Box

Type Series Booklet



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Type Series Booklet AmaDrainer Box

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Contents

Drainage	4
Lifting Units	4
AmaDrainer Box	4
Main applications.....	4
Fluids handled	4
Operating data.....	4
Design details	4
Designation	5
Materials.....	5
Product benefits.....	6
Product information	6
Certifications	6
Selection information	6
Overview of product features / selection tables	7
Characteristic curves.....	11
Dimensions and connections	15
Installation examples	19
Scope of supply	21
Control units.....	22
Accessories.....	25

Drainage

Lifting Units

AmaDrainer Box



Main applications

- Automatic drainage of buildings and premises below the flood level

Single-pump lifting units:

- Automatic disposal of domestic grey water from washing machines, washbasins, sinks, showers or bathtubs
- Disposal of highly contaminated, fibre-containing water from industrial applications, e.g. laundries, common washing facilities, washing machines and dish washers (also hot water cycle)
- Overflow protection for rainwater harvesting systems or inlet tanks of fire-fighting systems

Dual-pump lifting units:

- Design units requiring uninterrupted grey water drainage as dual-pump units (Z), as stipulated in EN 12 050-2.
- Automatic drainage of rooms, garage driveways and cellars (frost-free) with a leakage risk
- Overflow protection for rainwater harvesting systems or inlet tanks of fire-fighting systems

Fluids handled

AmaDrainer 301, 303 or AmaDrainer A / 10 / 11 - standard design for grey water:

- Slightly contaminated water
- Solid particles with a particle size of up to 10 or 11 mm

AmaDrainer 322 - standard design for grey water:

- Grey water containing fibres
- Solid particles with a particle size of up to 18 mm

AmaDrainer 354 or AmaDrainer A / 35 - standard design for grey water:

- Grey water containing long fibres and stringy material
- Solid particles with a particle size of up to 35 mm

Material variant C for aggressive grey water:

- Condensate from heat recovery applications
- Lab waste water containing low concentrations of foreign substances (pH 5 to 12)
- Cleaning agents, disinfecting agents, aggressive dishwashing agents and laundry agents (DIN 1986-3)
- Solid particles with a particle size of up to 35 mm

Material variant R for grey water containing oil / oil emulsions:

- Oil-water emulsions in car repair shops and at petrol stations
- Solid particles with a particle size of up to 10 or 11 mm

i AmaDrainer Box is not suitable for waste water from urinal and toilet systems. Use a MiniCompacta sewage lifting unit for this purpose.

Operating data

Table 1: Operating properties

Characteristic		Value	
		Single-pump station	Dual-pump station
Flow rate	Q [m³/h]	≤ 35	≤ 46
	Q [l/s]	≤ 9,7	≤ 12,8
Head	H [m]	≤ 21	≤ 24
Fluid temperature	T [°C]	≤ +40 (continuous)	
		≤ +90 (for short periods, ≤ 3 minutes)	
Total tank volume	V [l]	≤ 100	≤ 200

Design details

Design

- Single-pump unit or dual-pump unit
- 1 or 2 submersible motor pumps
- Fill level measurement via float switch
- Nominal tank volume 100 l or 200 l
- Collecting tank made of impact-resistant plastic
- Piping
- Integrated swing check valve
- To EN 12050-2
- Level-adjusting extension, cover plate with floor drain and siphon trap (underfloor box)
- Closed by odour-proof cover and locking ring (above-floor box)

Installation

- Installation in substructures or foundations (underfloor box)
- Floor mounting (above-floor box)

Drive

- AC motor
- Three-phase motor
- Integrated temperature switches

Impeller type

- Open multi-vane impeller, free passage 10 mm or 11 mm
- Vortex impeller, free passage 18 mm or 35 mm

Connections

- Inlet side DN 50 / DN 70 / DN 100 (single-pump lifting unit)
- Inlet side DN 50 / DN 70 / DN 100 / DN 150 (dual-pump lifting unit)
- Discharge outlet DN 50 / G 2

Designation
Example: AmaDrainer Box Z2.1 U 301
Table 2: Designation key

Code	Description	
AmaDrainer Box	Type series	
Z	Design	
	. ¹⁾	Single-pump lifting unit
	Z	Dual-pump lifting unit
2	Collecting tank volume	
	1	100 l
	2	200 l
.1	Version	
	.1	Version .1
U	Installation type	
	B	Above-floor installation
	U	Underfloor installation
301	Pump set	
	301	AmaDrainer 301

Materials
Table 3: Overview of available materials

Part No.	Description	Installation	
		Underfloor	Above-floor
591	Collecting tank	Polyethylene (PE)	Polyethylene (PE)
13-21	Extension	Polyethylene (PE)	-
185	Cover	Polyurethane (PUR)	-
160	Cover	-	Polyethylene (PE)
68-3	Floor drain	Polyamide (PA)	-
655	Pump	See AmaDrainer 3, 4 / 5 type series booklet	

¹ Blank

Product benefits

Reliable single-pump lifting unit / dual-pump lifting unit for underfloor installation / above-floor installation for all pumps of the AmaDrainer type series

Underfloor box U:

- Floor drain with siphon trap integrated in the cover plate
- Level-adjustable and rotatable extension
- Integrated swing check valve
- Broad range of submersible motor pumps
- Single-pump unit or dual-pump unit
- Graded-diameter nozzles offset by 180° for connecting inlet and vent lines
- Sealing collar for connecting a moisture barrier (optional)
- Puddle flange to protect against penetrating groundwater (optional)

Above-floor box B:

- Odour-proof cover (for Box 1 B incl. locking ring for tool-free fitting and removal)
- Integrated swing check valve
- Broad range of submersible motor pumps
- Single-pump unit or dual-pump unit
- Graded-diameter nozzles offset by 180° for connecting inlet and vent lines
- The discharge line can be connected either on the left- or right-hand side, as well as vertically (Box 1 B).

Lightning protection

- Electrical installations must be protected against overvoltage (binding since 14 December 2018) (see DIN VDE 0100-443 (IEC 60364-4-44:2007/A1:2015, modified) and DIN VDE 0100-534 (IEC 60364-5-53:2001/A2:2015, modified)). Whenever modifications are made to existing installations, retrofitting a surge protective device (SPD) in accordance with VDE is mandatory.
- A maximum cable length of 10 metres should not be exceeded between the surge protective device (usually type 1, internal lightning protection) installed at the service entrance and the equipment to be protected. For longer cables, additional surge protective devices (type 2) must be provided in the sub-distribution board upstream of the equipment to be protected or directly in the equipment itself.
- Sensor cables that cross the boundaries between lightning protection zones must be additionally protected by suitable surge protective devices (e.g. when using a 4 - 20 mA immersion probe).
- Control units installed outdoors (e.g. in outdoor cabinets) should always be equipped with Type 1 surge protective devices (lightning protection), as they will not normally be protected by a suitable upstream service entrance SPD.
- The associated lightning protection concept must be provided by the operator or by a suitable provider commissioned by the operator. Surge protective devices can be offered for the control units on request.

Product information

Product information as per Regulation No. 1907/2006 (REACH)

For information as per European chemicals regulation (EC) No. 1907/2006 (REACH) see <https://www.ksb.com/en-global/company/corporate-responsibility/reach>.

Certifications

Table 4: Overview

Label	Effective in:
 Bauart geprüft und überwacht www.tuv.com ID 1111220216	Europe

Selection information

Underfloor box installation

The underfloor box is suitable for floor slabs with a concrete thickness ≤ 25 cm.

For underfloor installation in waterproof concrete a puddle flange must be used. The installation must be compliant with DIN 18533. For single-layer sealing / water impact: W2.1-E, water exerting a constant hydrostatic pressure ≤ 3 m, make sure the sealing sheet is fitted properly.

For groundwater exerting a constant hydrostatic pressure > 3 m a grey water lifting unit designed for above-floor installation has to be used. The grey water lifting unit has to be installed in a "white tank" (waterproof concrete construction) with integrated recess.

Overview of product features / selection tables

Overview of fluids handled

The table of fluids handled is a selection aid for different applications. It serves as guidance and is based on long-standing experience. The data are reference values and are not to be considered generally binding recommendations. They shall not be the basis for warranty claims. Please contact your nearest sales branch for in-depth advice.

Table 5: Table of fluids handled

Fluids handled	pH	AmaDrainer 3			AmaDrainer 4 / 5		
		Free passage					
		10	18	35	10	11	35
		[mm]	[mm]	[mm]	[mm]	[mm]	[mm]
Slightly contaminated water	-	X	X	X	X	X	X
Grey water containing fibres	-	-	X	X	-	-	X
Grey water containing long fibres and stringy material	-	-	-	X	-	-	X
Condensate from heat recovery applications	-	X ²⁾	X ²⁾	X ²⁾	X ²⁾	X ²⁾	-
Waste water from laboratories	5 to 12	X ²⁾	X ²⁾	X ²⁾	X ²⁾	X ²⁾	-
Oil-water emulsion	-	-	-	-	X ³⁾	X ³⁾	-
Cleaning lye / washing lye	< 12	X ²⁾	X ²⁾	X ²⁾	X ²⁾	X ²⁾	-

²⁾ Variant C

³⁾ Variant R

Off-the-shelf range

Table 6: Technical data of the grey water lifting unit, off-the-shelf (underfloor installation)

Grey water lifting unit			Submersible grey water pump					Mat. No.	[kg]
Single-pump lifting unit ⁴⁾	Dual-pump lifting unit	Total volume	Size	P ₂	I _N 1~ 230 V, 50 Hz	Power cable			
		[l]		[kW]		[A]	[m]		
Free passage: 10 mm									
1.1 U 301	-	100	301	0,25	1,90	10	3 × 0,75 ⁵⁾	29135141	20.3
-	Z2.1 U 301	200	301	0,25	1,90	10	3 × 0,75 ⁵⁾	29135142	49.7
1.1 U 303	-	100	303	0,60	4,00	10	3 × 0,75 ⁵⁾	29135143	21.5
-	Z2.1 U 303	200	303	0,60	4,00	10	3 × 0,75 ⁵⁾	29135144	52.1
Free passage 18 mm									
1.1 U 322	-	100	322	0,40	3,50	10	3 × 0,75 ⁵⁾	29135145	21.5
-	Z2.1 U 322	200	322	0,40	3,50	10	3 × 0,75 ⁵⁾	29135146	52.1
Free passage 35 mm									
1.1 U 354	-	100	354	0,60	4,00	10	3 × 0,75 ⁵⁾	29135147	21.8
-	Z2.1 U 354	200	354	0,60	4,00	10	3 × 0,75 ⁵⁾	29135148	52.5
	-						6)		
-							6)		

Table 7: Technical data of the grey water lifting unit, off-the-shelf (above-floor installation)

Grey water lifting unit			Submersible grey water pump					Mat. No.	[kg]
Single-pump lifting unit ⁴⁾	Dual-pump lifting unit	Total volume	Size	P ₂	I _N 1~ 230 V, 50 Hz	Power cable			
		[l]		[kW]		[A]	[m]		
Free passage: 10 mm									
1.1 B 301	-	100	301	0,25	1,90	10	3 × 0,75 ⁵⁾	29135149	13.3
-	Z2.1 B 301	200	301	0,25	1,90	10	3 × 0,75 ⁵⁾	29135150	47.3
1.1 B 303	-	100	303	0,60	4,00	10	3 × 0,75 ⁵⁾	29135151	14.5
-	Z2.1 B 303	200	303	0,60	4,00	10	3 × 0,75 ⁵⁾	29135152	49.7
Free passage 18 mm									
1.1 B 322	-	100	322	0,40	3,50	10	3 × 0,75 ⁵⁾	29135153	14.5
-	Z2.1 B 322	200	322	0,40	3,50	10	3 × 0,75 ⁵⁾	29135154	49.7
Free passage 35 mm									
1.1 B 354	-	100	354	0,60	4,00	10	3 × 0,75 ⁵⁾	29135155	15
-	Z2.1 B 354	200	354	0,60	4,00	10	3 × 0,75 ⁵⁾	29135156	50.1
1.1 B 405	-	100	A 405 SE/35	0,55	4,10	10	3 × 1,0 ⁶⁾	29131766	23.9
-	Z2.1 B 405	200	A 405 NE/35	0,55	4,10	10	3 × 1,0 ⁶⁾	29131767	65.3

⁴⁾ Single-pump units in conjunction with LevelControl Basic 2 are optionally available from the configurable range only.

⁵⁾ Power cable H05RN8-F

⁶⁾ Power cable H07RN-FG

Configurable range of single-pump units

i For aggressive grey water use a pump set of material variant C; for grey water containing oil use material variant R. See type series booklet of submersible grey water pump or KSB EasySelect.

i AmaDrainer 4 / 5: If a LevelControl Basic 2 control unit is to be used, a pump set without float switch (version NE/ND) is required.

Table 8: Configurable range of single-pump units

	AmaDrainer Box	P ₂	I _N		Power cable		Mat. No.	[kg]
			1~220-240 V, 50 Hz	3~380-415 V, 50 Hz				
		[kW]	[A]	[A]	[m]	[mm ²]		
1	Collecting tank 1.1 U - 100 l with level-adjusting extension and cover with floor drain (underfloor)						19074492	15.7
	Collecting tank 1.1 B - 100 l with cover and locking ring (above floor)						19074493	8.7
2	AmaDrainer DN 32, free passage 10 mm (underfloor / above-floor)							
	301	0,25	1,90	-	10	3 × 0,75 ⁷⁾	48267549	4.2
	303	0,60	4,00	-	10	3 × 0,75 ⁷⁾	48267550	5.5
	AmaDrainer DN 40, free passage 10 mm (underfloor / above-floor)							
	A 405 SE/10	0,55	3,50	-	10	3 × 1.0 ⁸⁾	29128650	12.7
	A 405 SD/10	0,55	-	1,70	10	6 × 1.0 ⁸⁾	29128742	15.1
	A 407 SE/10	0,75	4,50	-	10	3 × 1.0 ⁸⁾	29128653	12.7
	A 407 SD/10	0,75	-	1,90	10	6 × 1.0 ⁸⁾	29128743	15.1
	A 411 SE/10	1,10	6,85	-	10	3 × 1.0 ⁸⁾	29128656	15
	A 411 SD/10	1,10	-	2,50	10	6 × 1.0 ⁸⁾	29128744	15.1
	A 415 SE/10	1,50	9,60	-	10	3 × 1.0 ⁸⁾	29128659	15
	A 415 SD/10	1,50	-	3,60	10	6 × 1.0 ⁸⁾	29128745	16.9
	A 422 SD/10	2,20	-	4,80	10	6 × 1.0 ⁸⁾	29128746	16.9
	AmaDrainer DN 50, free passage 10 mm (with cooling jacket, underfloor / above-floor)							
	A 505 SE/10K	0,55	3,50	-	10	3 × 1.0 ⁸⁾	29128663	14.7
	A 505 SD/10K	0,55	-	1,70	10	6 × 1.0 ⁸⁾	29128747	17.1
	A 507 SE/10K	0,75	4,50	-	10	3 × 1.0 ⁸⁾	29128666	14.7
	A 507 SD/10K	0,75	-	1,90	10	6 × 1.0 ⁸⁾	29128748	17.1
	A 511 SE/10K	1,10	6,85	-	10	3 × 1.0 ⁸⁾	29128669	17
	A 511 SD/10K	1,10	-	2,50	10	6 × 1.0 ⁸⁾	29128749	17.1
	A 515 SE/10K	1,50	9,60	-	10	3 × 1.0 ⁸⁾	29128672	17
	A 515 SD/10K	1,50	-	3,60	10	6 × 1.0 ⁸⁾	29128750	18.9
	A 522 SD/10K	2,20	-	4,80	10	6 × 1.0 ⁸⁾	29128751	18.9
	AmaDrainer DN 32, free passage 18 mm (underfloor / above-floor)							
	322	0,40	3,50	-	10	3 × 0,75 ⁷⁾	48267551	5.5
	AmaDrainer DN 40, free passage 35 mm (underfloor / above-floor)							
	354	0,60	4,00	-	10	3 × 0,75 ⁷⁾	48267552	5.6
	A 405 SE/35	0,55	3,50	-	10	3 × 1.0 ⁸⁾	29128676	13.7
	A 405 SD/35	0,55	-	1,70	10	6 × 1.0 ⁸⁾	29128752	16.1
	A 411 SE/35	1,10	6,85	-	10	3 × 1.0 ⁸⁾	29128679	16
	A 411 SD/35	1,10	-	2,50	10	6 × 1.0 ⁸⁾	29128753	16.1
	A 422 SD/35	2,20	-	4,80	10	6 × 1.0 ⁸⁾	29128754	17.9
3	Connection set with full set of installation parts (underfloor / above-floor)							
	AmaDrainer 301, DN 32 - free passage 10 mm						19066525	0.4
	AmaDrainer 303, DN 32 - free passage 10 mm						19074496	0.4
	AmaDrainer 322, DN 32 - free passage 18 mm						19066526	0.4
	AmaDrainer 4 / 5, DN 40 - free passage 10 mm						19074497	0.4
	AmaDrainer 354, DN 40 - free passage 35 mm						19066527	0.7
	AmaDrainer 4 / 5, DN 40 - free passage 35 mm						19074506	1
	AmaDrainer 4 / 5, DN 50 - free passage 10 mm (with cooling jacket)						19074498	0.4

⁷ Power cable H05RN8-F

⁸ Power cable H07RN-FG

Configurable range of dual-pump units

i For aggressive grey water use a pump set of material variant C; for grey water containing oil use material variant R. See type series booklet of submersible grey water pump or KSB EasySelect.

i Oil-resistant float switches (PUR) are required for material variant R. (⇒ Page 27)

Table 9: Configurable range of dual-pump units

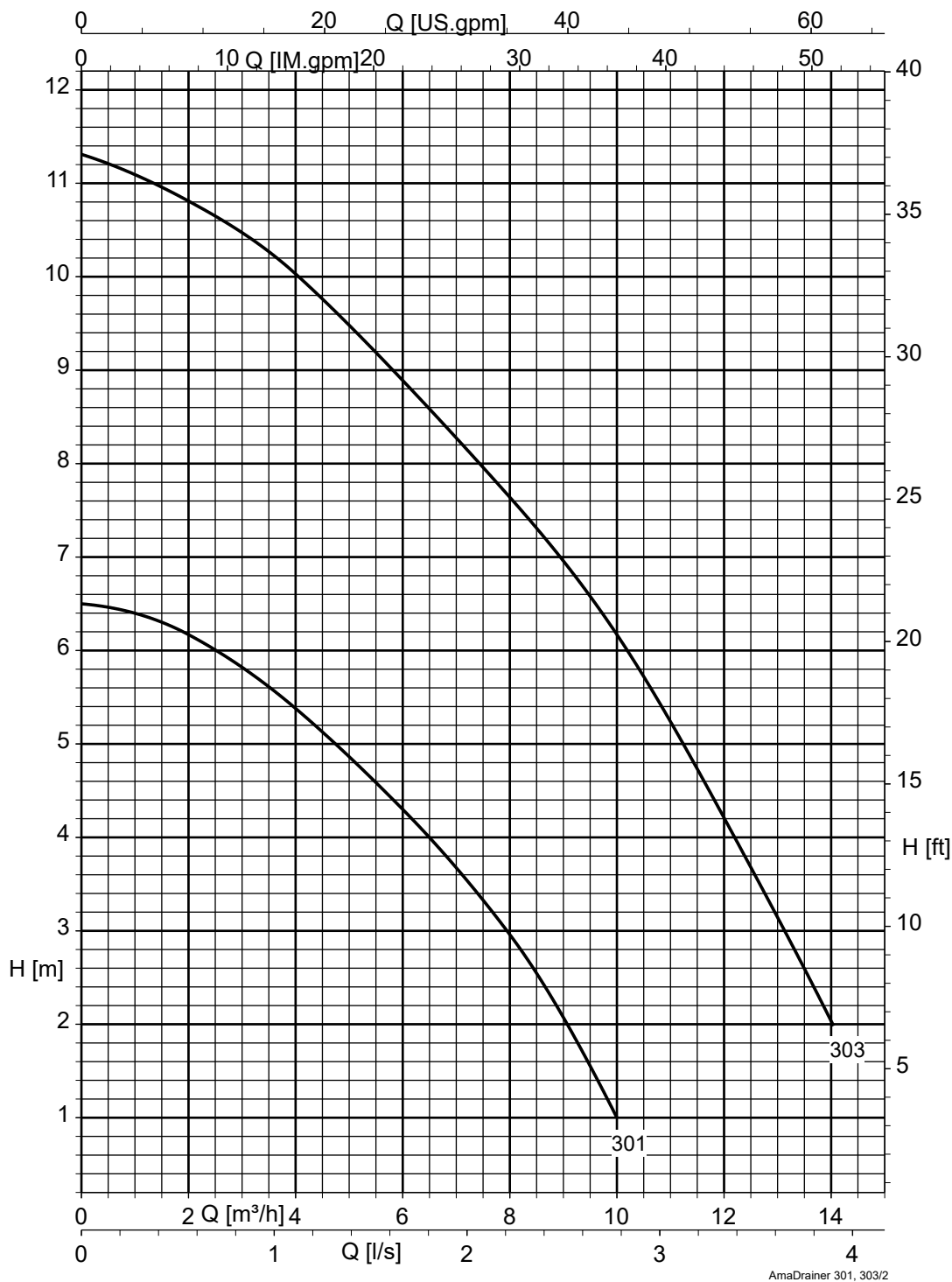
	AmaDrainer Box	P ₂	I _N		Power cable		Mat. No.	[kg]
			1~220-240 V, 50 Hz	3~380-415 V, 50 Hz				
		[kW]	[A]	[A]	[m]	[mm ²]		
1	Collecting tank Z2.1 U - 200 l with level-adjusting extension and cover with floor drain (underfloor)						19074494	31.3
	Collecting tank Z2.1 B - 200 l with cover (above floor)						19074495	28.9
2	AmaDrainer DN 32, free passage 10 mm (underfloor / above-floor)							
	301	0,25	1,90	-	10	3 × 0,75 ⁹⁾	48267549	4.2
	303	0,60	4,00	-	10	3 × 0,75 ⁹⁾	48267550	5.5
	AmaDrainer DN 40, free passage 10 mm (underfloor / above-floor)							
	A 405 NE/10	0,55	3,50	-	10	3 × 1.0 ¹⁰⁾	29128651	12.2
	A 405 ND/10	0,55	-	1,70	10	6 × 1.0 ¹⁰⁾	29128652	13.8
	A 407 NE/10	0,75	4,50	-	10	3 × 1.0 ¹⁰⁾	29128654	12.2
	A 407 ND/10	0,75	-	1,90	10	6 × 1.0 ¹⁰⁾	29128655	13.8
	A 411 NE/10	1,10	6,85	-	10	3 × 1.0 ¹⁰⁾	29128657	14.5
	A 411 ND/10	1,10	-	2,50	10	6 × 1.0 ¹⁰⁾	29128658	13.8
	A 415 NE/10	1,50	9,60	-	10	3 × 1.0 ¹⁰⁾	29128660	14.5
	A 415 ND/10	1,50	-	3,60	10	6 × 1.0 ¹⁰⁾	29128661	15.6
	A 422 ND/10	2,20	-	4,80	10	6 × 1.0 ¹⁰⁾	29128662	15.6
	A 522 ND/11	2,20	-	4,80	10	6 × 1.0 ¹⁰⁾	29128865	25
	AmaDrainer DN 50, free passage 10 mm (with cooling jacket, underfloor / above-floor)							
	A 505 NE/10K	0,55	3,50	-	10	3 × 1.0 ¹⁰⁾	29128664	14.2
	A 505 ND/10K	0,55	-	1,70	10	6 × 1.0 ¹⁰⁾	29128665	15.8
	A 507 NE/10K	0,75	4,50	-	10	3 × 1.0 ¹⁰⁾	29128667	14.2
	A 507 ND/10K	0,75	-	1,90	10	6 × 1.0 ¹⁰⁾	29128668	15.8
	A 511 NE/10K	1,10	6,85	-	10	3 × 1.0 ¹⁰⁾	29128670	16.5
	A 511 ND/10K	1,10	-	2,50	10	6 × 1.0 ¹⁰⁾	29128671	15.8
	A 515 NE/10K	1,50	9,60	-	10	3 × 1.0 ¹⁰⁾	29128673	16.5
	A 515 ND/10K	1,50	-	3,60	10	6 × 1.0 ¹⁰⁾	29128674	17.6
	A 522 ND/10K	2,20	-	4,80	10	6 × 1.0 ¹⁰⁾	29128675	17.6
	AmaDrainer DN 32, free passage 18 mm (underfloor / above-floor)							
	322	0,40	3,50	-	10	3 × 0,75 ⁹⁾	48267551	5.5
	AmaDrainer DN 40, free passage 35 mm (underfloor / above-floor)							
	354	0,60	4,00	-	10	3 × 0,75 ⁹⁾	48267552	5.6
	A 405 NE/35	0,55	3,50	-	10	3 × 1.0 ¹⁰⁾	29128677	13.2
	A 405 ND/35	0,55	-	1,70	10	6 × 1.0 ¹⁰⁾	29128678	14.8
	A 411 NE/35	1,10	6,85	-	10	3 × 1.0 ¹⁰⁾	29128680	15.5
	A 411 ND/35	1,10	-	2,50	10	6 × 1.0 ¹⁰⁾	29128681	14.8
	A 422 ND/35	2,20	-	4,80	10	6 × 1.0 ¹⁰⁾	29128682	16.6
3	Connection set with full set of installation parts and 2 float switches - power cable H07, cable length 10 m (underfloor / above-floor)							
	AmaDrainer 301, DN 32 - free passage 10 mm						19066521	5.5
	AmaDrainer 303, DN 32 - free passage 10 mm						19066522	5.5
	AmaDrainer 322, DN 32 - free passage 18 mm						19066523	5.5
	AmaDrainer 4 / 5, DN 40 - free passage 10 mm						19066517	5.8
	AmaDrainer 4 / 5, DN 50 - free passage 11 mm						19066518	5.8
	AmaDrainer 4 / 5, DN 50 - free passage 10 mm (with cooling jacket)						19066519	5.5
	AmaDrainer 354, DN 40 - free passage 35 mm						19066524	5.5
	AmaDrainer 4 / 5, DN 40 - free passage 35 mm						19066520	5.8

⁹⁾ Power cable H05RN8-F

¹⁰⁾ Power cable H07RN-FG

Characteristic curves

AmaDrainer 301, 303, $n = 2800$ rpm

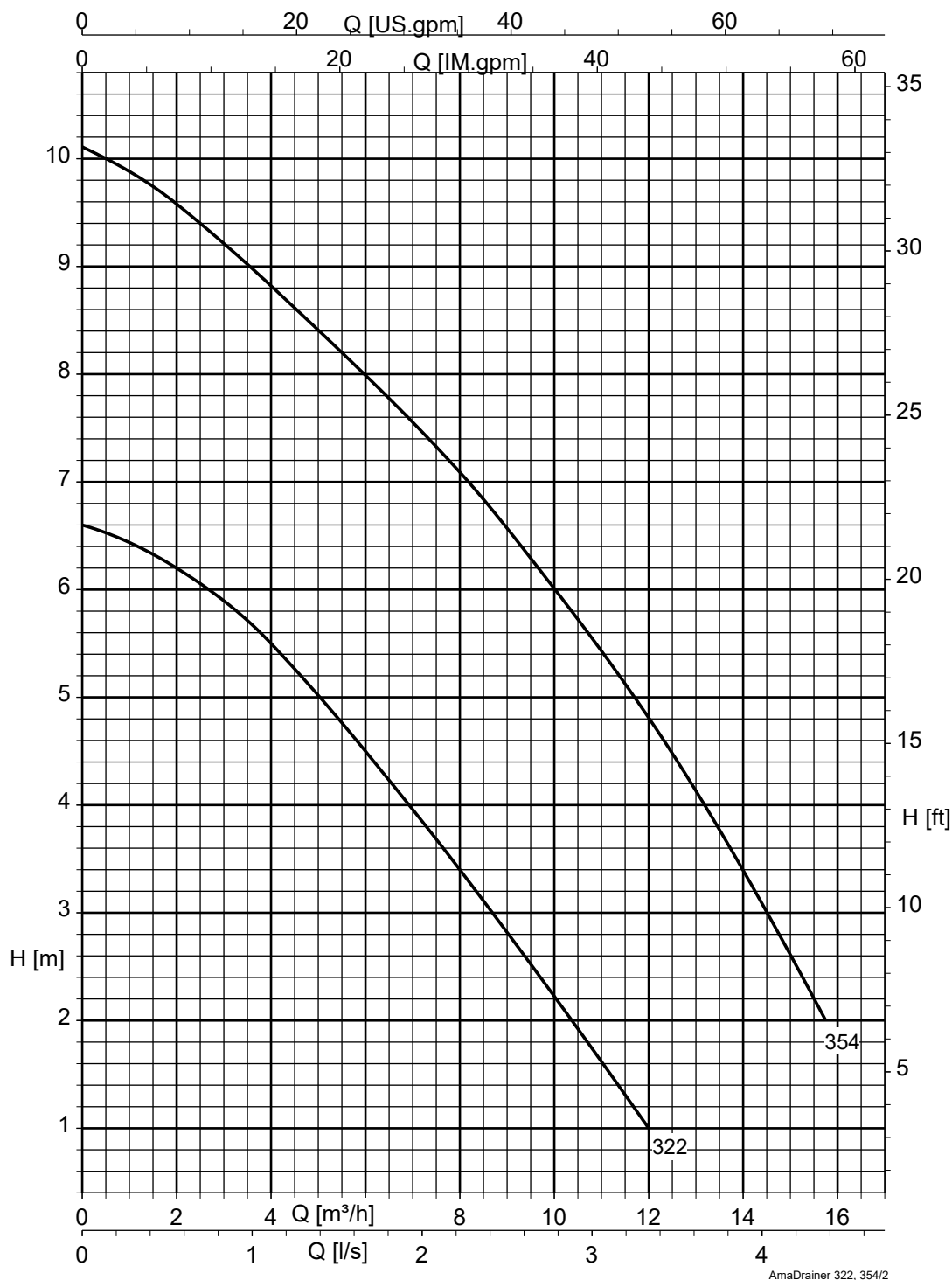


Sizes 301, 303: free passage = 10 mm

Performance tolerance to ISO 9906, Annex A (water under standard conditions)

The characteristic curves shown are those of the respective pumps. Hydraulic losses caused by the discharge-side piping, both inside and outside of the AmaDrainer Box lifting unit, must be taken into account during product selection.

AmaDrainer 322, 354, n = 2800 rpm



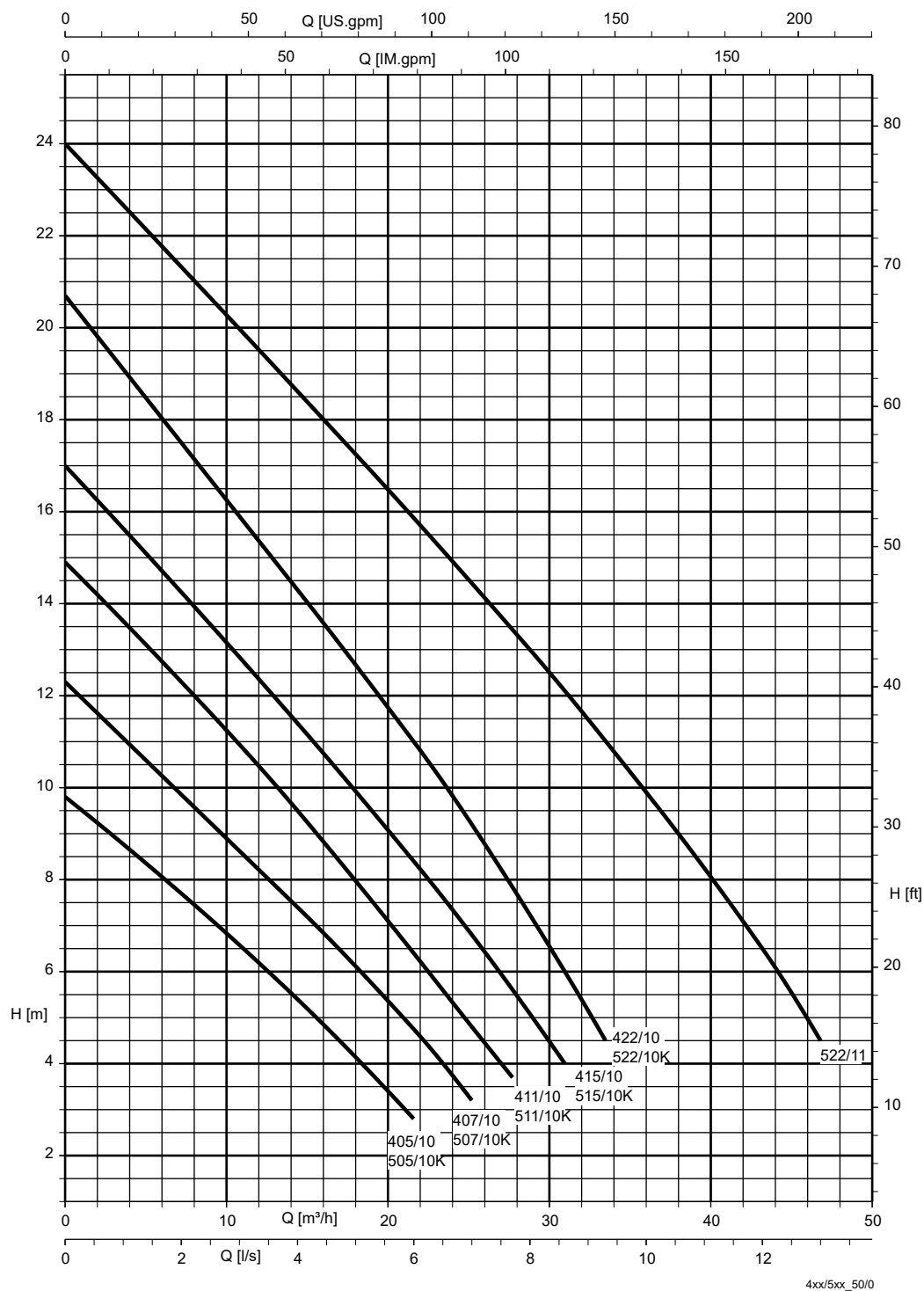
Size 322: free passage = 18 mm

Size 354: free passage = 35 mm

Performance tolerance to ISO 9906, Annex A (water under standard conditions)

The characteristic curves shown are those of the respective pumps. Hydraulic losses caused by the discharge-side piping, both inside and outside of the AmaDrainer Box lifting unit, must be taken into account during product selection.

AmaDrainer 4 / 5, n = 2800 rpm, multi-vane impeller



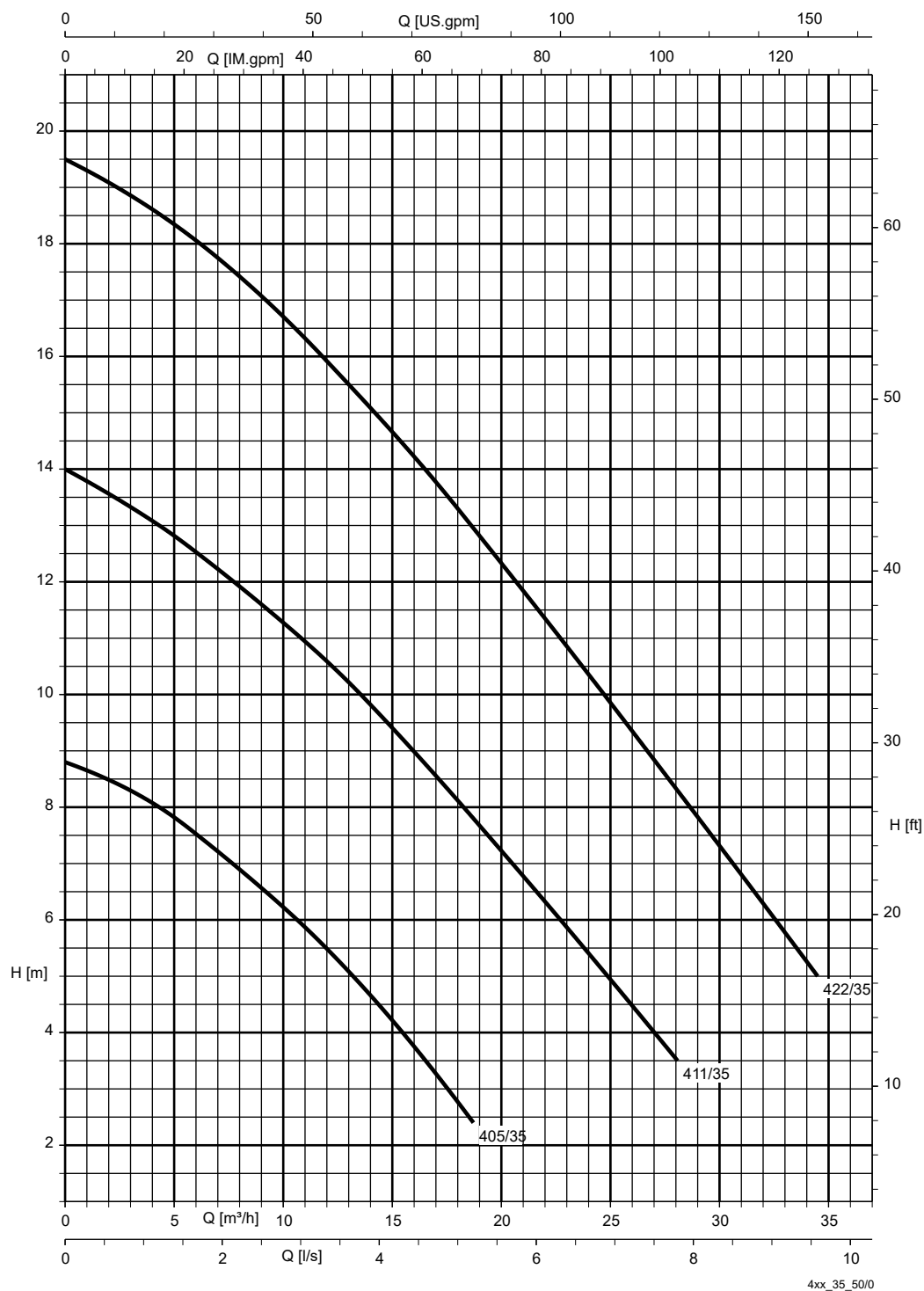
Sizes 405, 407, 411, 415, 422, 505, 507, 511, 515: free passage = 10 mm

Size 522: free passage = 11 mm

Performance tolerance to ISO 2548 Class C (water under standard conditions)

The characteristic curves shown are those of the respective pumps. Hydraulic losses caused by the discharge-side piping, both inside and outside of the AmaDrainer Box lifting unit, must be taken into account during product selection.

AmaDrainer 4, n = 2800 rpm, F impeller



Sizes 405, 411, 422: free passage = 35 mm

Performance tolerance to ISO 2548 Class C (water under standard conditions)

The characteristic curves shown are those of the respective pumps. Hydraulic losses caused by the discharge-side piping, both inside and outside of the AmaDrainer Box lifting unit, must be taken into account during product selection.

Dimensions and connections

Underfloor box: Dimensions and connections

Single-pump lifting unit

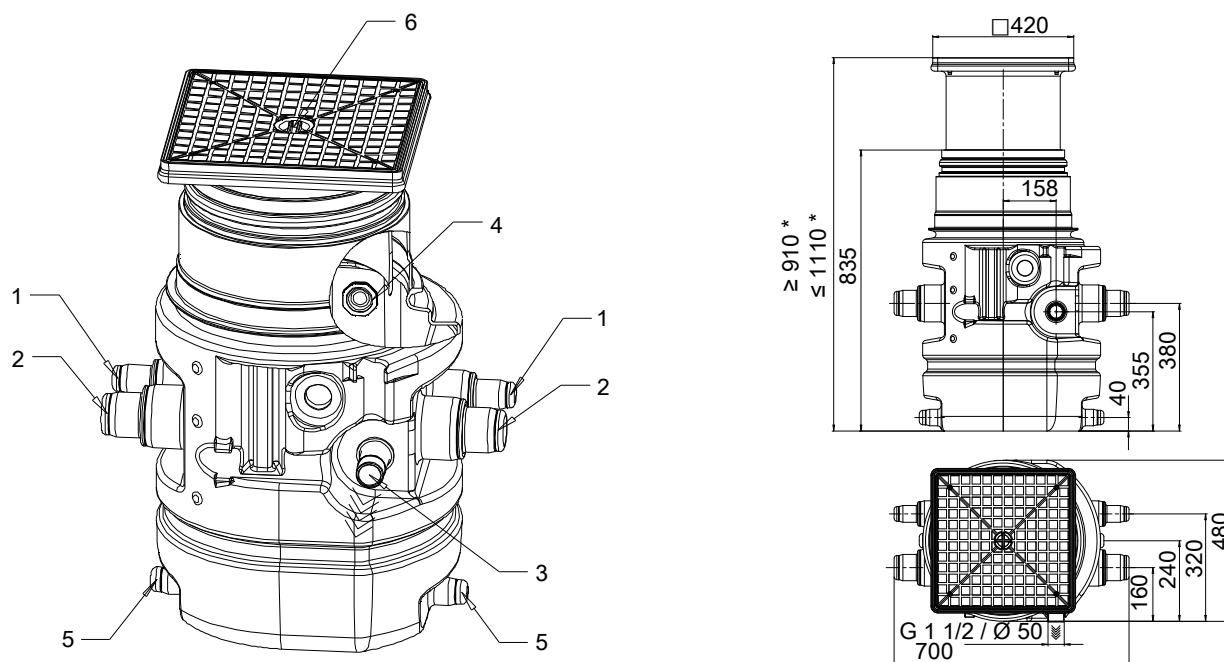


Fig. 1: Connections and dimensions of a single-pump lifting unit

* Installation depth

Table 10: Overview of connections

Position	Connection	Description
1	Inlet	2× connection nozzle DN 50 / DN 70 graded, horizontal, opposite each other Outside diameter 50 / 78 mm
2	Inlet	2× connection nozzle DN 70 / DN 100 graded, horizontal, opposite each other Outside diameter 78 / 110 mm
1, 2	Cable entry / venting	Free connection nozzle DN 50 / DN 70 / DN 100
3	Discharge-side connection	DN 40 horizontal, threaded connection G 1 1/2 / smooth pipe end (glued nozzle), outside diameter 50 mm
4	Cable gland	PG 36 for conduit Inside diameter 24 mm (for single-phase AC motor only)
5	Emergency drainage	2× DN 40 horizontal, opposite each other Outside diameter 49 mm
6	Floor drain	DN 50

Dual-pump lifting unit

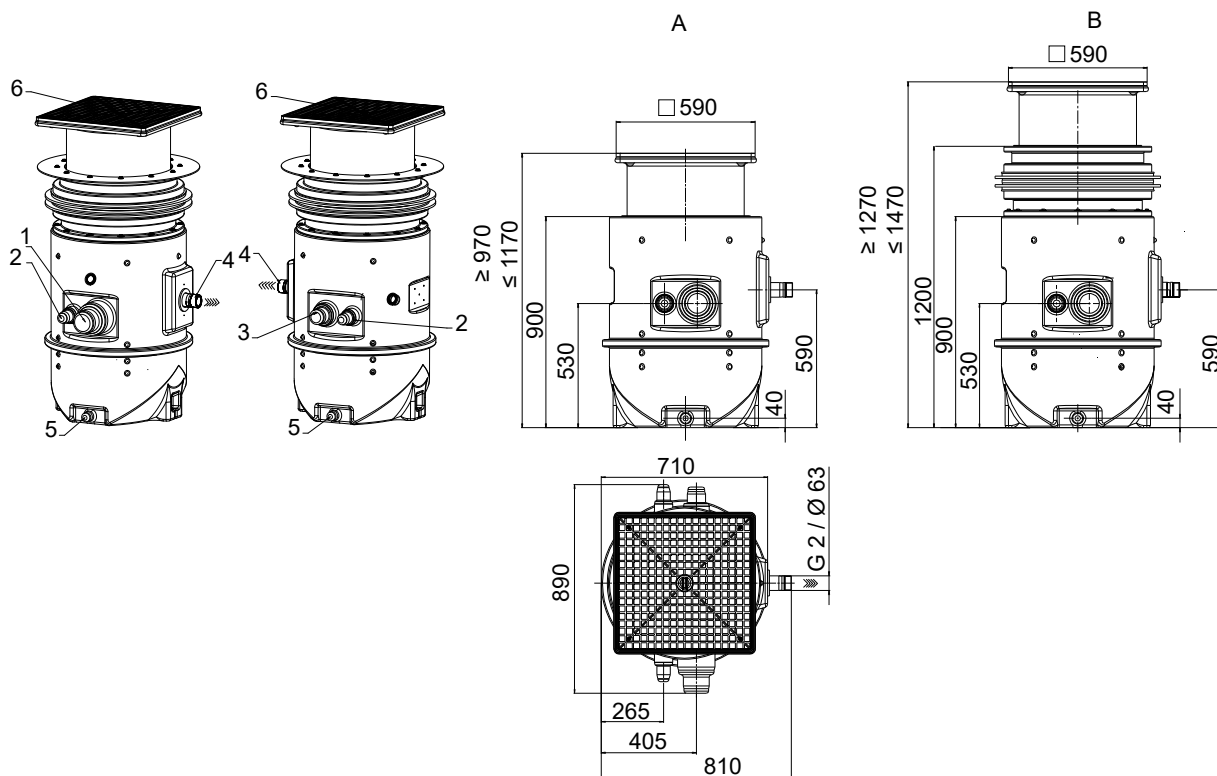


Fig. 2: Connections and dimensions of a dual-pump lifting unit

A	Standard design	B	With level-adjusting piece
*	Installation depth		

Table 11: Overview of connections

Position	Connection	Description
1	Inlet	1× connection nozzle DN 100 / DN 150 graded, horizontal Outside diameter 110 / 159 mm
2	Inlet	2× connection nozzle DN 50 / DN 70 graded, horizontal, opposite each other Outside diameter 58 / 78 mm
3	Inlet	1× connection nozzle DN 70 / DN 100 graded, horizontal Outside diameter 78 / 110 mm
1, 2, 3	Cable entry / venting	Free connection nozzle DN 50 / DN 70 / DN 100 / DN 150
4	Discharge-side connection	DN 50 horizontal, threaded connection G 2 / smooth pipe end (glued nozzle) Outside diameter 63 mm
5	Emergency drainage	2× DN 40 horizontal, opposite each other Outside diameter 49 mm
6	Floor drain	DN 50

Above-floor box: Dimensions and connections

Single-pump lifting unit

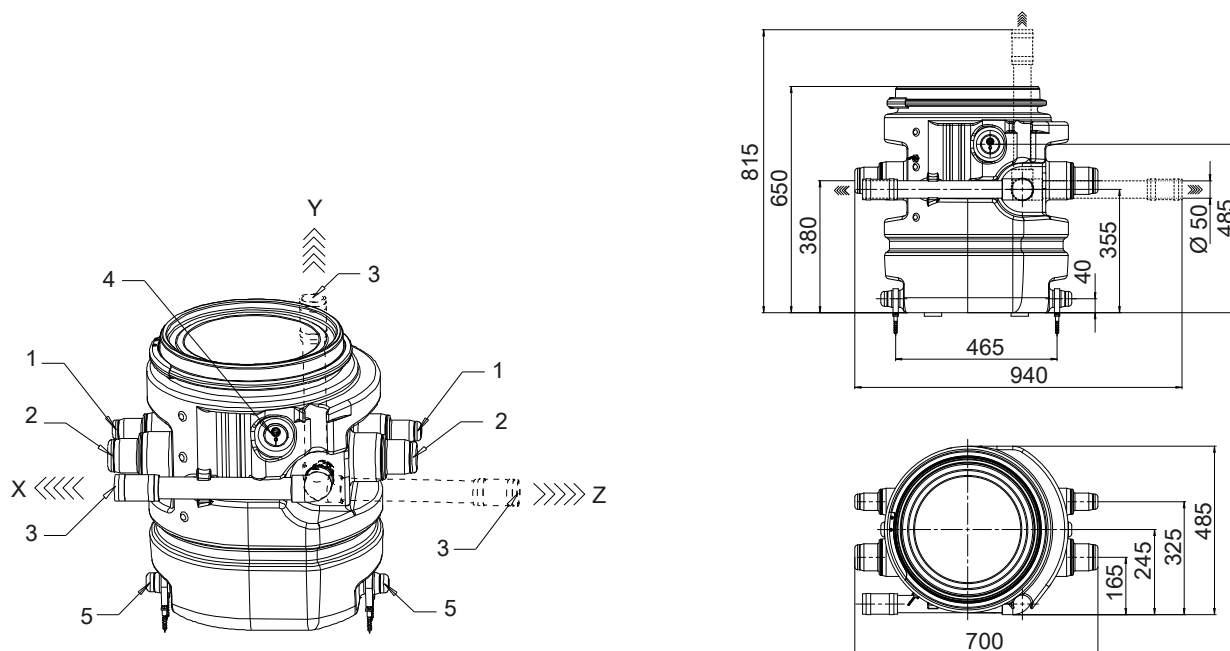


Fig. 3: Connections and dimensions of a single-pump lifting unit

Table 12: Overview of connections

Position	Connection	Description
1	Inlet	2x connection nozzle DN 50 / DN 70 graded, horizontal, opposite each other Outside diameter 50 / 78 mm
2	Inlet	2x connection nozzle DN 70 / DN 100 graded, horizontal, opposite each other Outside diameter 78 / 110 mm
1, 2	Vent	Free connection nozzle DN 50 / DN 70 / DN 100
3 X	Discharge-side connection	DN 40, horizontal on the right, with flexible hose connection Inside hose diameter 50 mm
3 Y	Discharge-side connection	DN 40, vertical, with flexible hose connection Inside hose diameter 50 mm
3 Z	Discharge-side connection	DN 40, horizontal on the left, with flexible hose connection Inside hose diameter 50 mm
4	Cable entry	Cable grommet A/B/C, depending on the cable diameter
5	Emergency drainage	2x DN 40 horizontal, opposite each other Outside diameter 49 mm

Dual-pump lifting unit

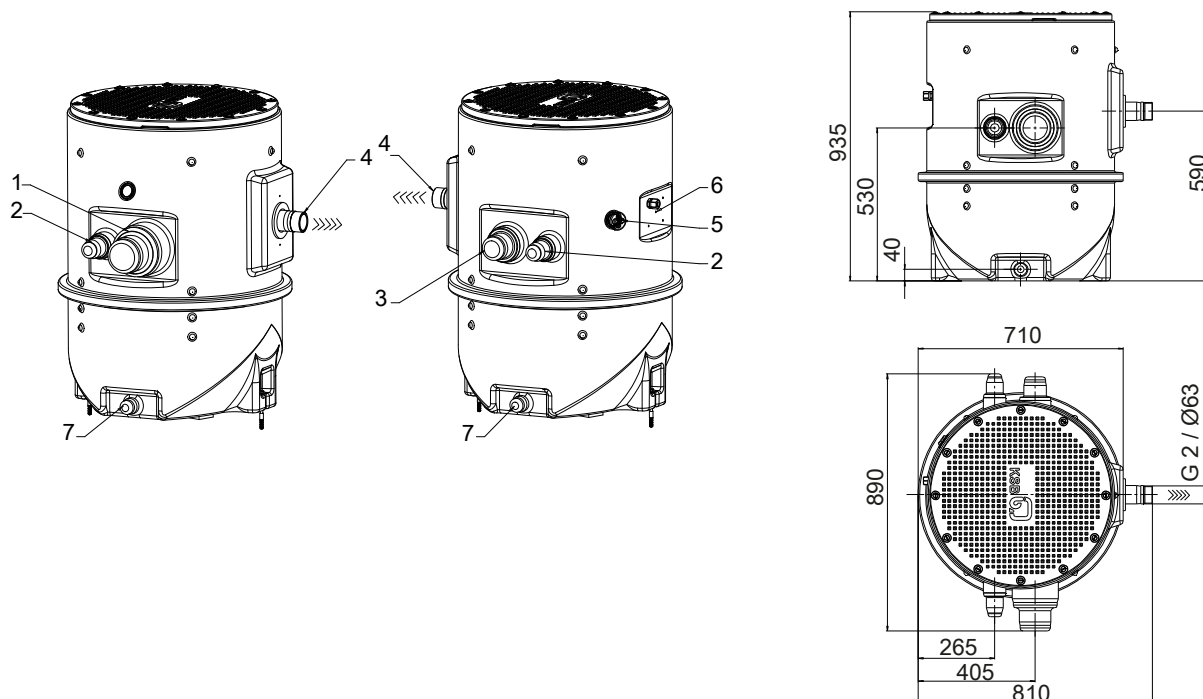


Fig. 4: Connections and dimensions of a dual-pump lifting unit

Table 13: Overview of connections

Position	Connection	Description
1	Inlet	1× connection nozzle DN 100 / DN 150 graded, horizontal Outside diameter 110 / 159 mm
2	Inlet	2× connection nozzle DN 50 / DN 70 graded, horizontal, opposite each other Outside diameter 58 / 78 mm
3	Inlet	1× connection nozzle DN 70 / DN 100 graded, horizontal Outside diameter 78 / 110 mm
1, 2, 3	Vent	Free connection nozzle DN 50 / DN 70 / DN 100 / DN 150
4	Discharge-side connection	DN 50, horizontal, with flexible hose connection Inside hose diameter 63 mm
5	Cable entry	Grommet for two motor power cables
6	Connector panel	For cable glands of float switches 5 × M20 (drill diameter 20 mm)
7	Emergency drainage	2× DN 40 horizontal, opposite each other Outside diameter 49 mm

Installation examples

Underfloor box: installation example

Single-pump lifting unit

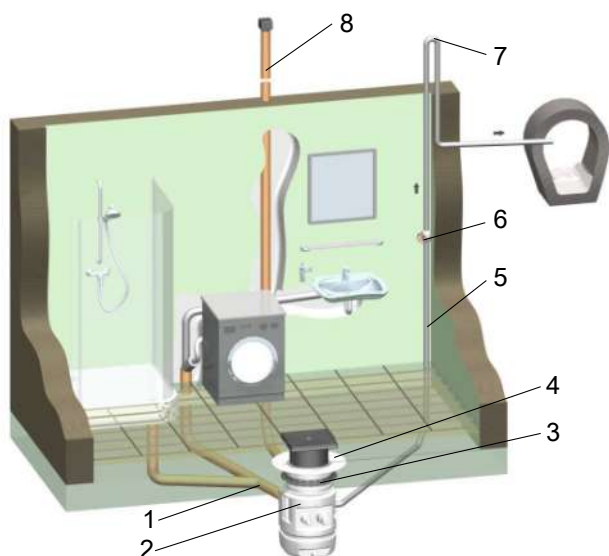


Fig. 5: Installation example

1	Inlet	2	Lifting unit
3	Puddle flange	4	Sealing collar
5	Discharge line	6	Socket gate valve
7	Flood level	8	Vent line

Dual-pump lifting unit

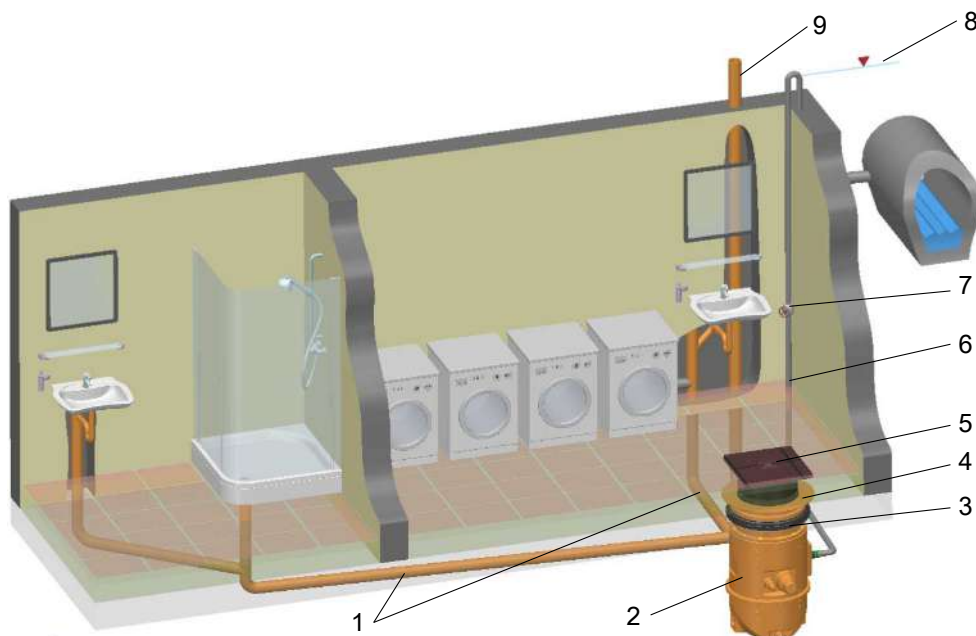


Fig. 6: Installation example

1	Inlet	2	Lifting unit
3	Puddle flange	4	Sealing collar
5	Floor drain	6	Discharge line
7	Socket gate valve	8	Flood level
9	Vent line		

Above-floor box: Installation example

Single-pump lifting unit

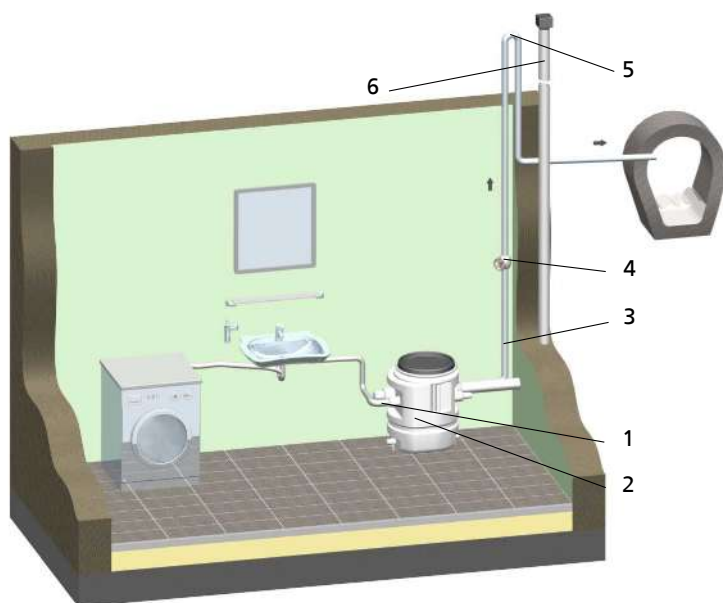


Fig. 7: Installation example

1	Inlet	2	Lifting unit
3	Discharge line	4	Socket gate valve
5	Flood level	6	Vent line

Dual-pump lifting unit

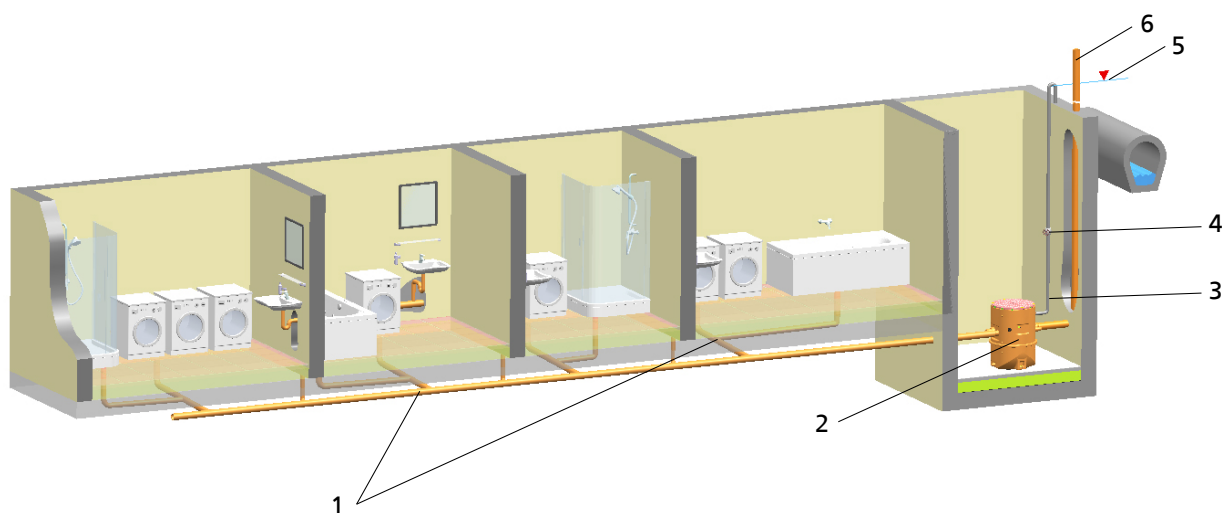


Fig. 8: Installation example

1	Inlet	2	Lifting unit
3	Discharge line	4	Socket gate valve
5	Flood level	6	Vent line

Scope of supply

The off-the-shelf range contains the complete grey water lifting unit as one material number.

The configurable range serves to individually select the components composing the grey water lifting unit.

In either case, the grey water lifting unit is supplied in several parts for assembly on site. The grey water lifting unit meets the requirements of EN 12050-2.

- Collecting tank with cover, locking ring
- Two submersible grey water pumps with 10 m power cable, shockproof plug or free cable end (for models with three-phase motor).
- Connection set with full set of installation parts, swing check valve¹⁶⁾, two float switches, with 10 m electric cable
- LevelControl Basic 2 control unit for dual-pump units with automatic pump changeover function, peak-load operation function, stand-by function

Underfloor box 1.1 U - single-pump unit

- Collecting tank made of impact-resistant plastic, 100 litres, for installation in substructures or foundations. Cover with level-adjusting extension, cover plate with floor drain and siphon trap.

Supplied in three parts:

- Collecting tank with extension, cover plate Class K3
- Submersible grey water pump that starts and stops automatically, with 10 m power cable, shockproof plug or CEE plug¹¹⁾
- Connection set with full set of installation parts, swing check valve¹²⁾

Underfloor Box Z2.1 U – dual-pump unit

- Collecting tank made of impact-resistant plastic, 200 litres, for installation in substructures and foundations. Cover with level-adjusting extension, cover plate with floor drain and siphon trap.

Supplied in four parts:

- Collecting tank with extension, cover plate Class K3
- Two submersible grey water pumps with 10 m power cable, shockproof plug or free cable end (for models with three-phase motor).
- Connection set with full set of installation parts, swing check valve¹³⁾, two float switches, with 10 m electric cable
- LevelControl Basic 2 control unit for dual-pump units with automatic pump changeover function, peak-load operation function, stand-by function

Above-floor box 1.1 B – single-pump unit

- Collecting tank made of impact-resistant plastic, 100 litres, for floor mounting; closed with odour-proof cover and locking ring

Supplied in three parts:

- Collecting tank with cover, locking ring
- Submersible grey water pump that starts and stops automatically, with 10 m power cable, shockproof plug or CEE plug¹⁴⁾
- Connection set with full set of installation parts, swing check valve¹⁵⁾

Above-floor box Z2.1 B – dual-pump unit

- Collecting tank made of impact-resistant plastic, 200 litres, for floor mounting; closed with odour-proof, screw-on cover

Supplied in four parts:

¹¹⁾ For models with three-phase motor
¹²⁾ AmaDrainer 301, 303 and 322 with integrated swing check valve
¹³⁾ AmaDrainer 301, 303 and 322 with integrated swing check valve
¹⁴⁾ For models with three-phase motor
¹⁵⁾ AmaDrainer 301, 303 and 322 with integrated swing check valve
¹⁶⁾ AmaDrainer 301, 303 and 322 with integrated swing check valve

Control units

Table 14: Overview of control units included in the scope of supply

Range	Single-pump lifting unit	Dual-pump lifting unit
Off-the-shelf range	-	X
Configurable range	-	-

i For dual-pump lifting units selected via the configurable range the required control unit has to be ordered separately. For single-pump lifting units an optional control unit is only available via the configurable range.

The control unit is supplied in enclosure IP54 and must be installed in a well-ventilated, flood-proof room.

LevelControl Basic 2 product description



LevelControl Basic 2 BC

Description

- Mains connection: 230 V or 400 V
- Numerical display with status indication (traffic light) and navigation keys
- Fill level display and operating data display
- Signal lamp per pump
- Indicator lamp for high water
- Manual-0-automatic selector switch
- Integrated alarm buzzer
- Pump protection:
 - With thermal circuit breaker
 - With circuit breaker (230 V)
 - With motor protection switch (400 V)
- Even distribution of pump operating hours due to automatic pump changeover (for dual-pump units)
- Parameterisable service intervals
- Diagnostic function and signalling/message function
- Intelligent voltage monitoring and phase monitoring (for 400 V: phase failure and phase sequence)
- Two inputs for external fault message and remote acknowledgement
- Volt-free general fault message (changeover contact)
- Numerous additional functions (e.g. monitoring of supply voltage, intelligent system monitoring, and many more)
- Optional: battery-backed mains-independent alarm
- Optional: integrated master switch
- Optional: signalling module with six volt-free individual relays

Control units for single-pump units from the configurable range

Table 15: Control units for single-pump units from the configurable range

Item	Description	Type	Current	AmaDrainer Box							Mat. No.	[kg]	
				1~	3~								
			[A]	...E	05 D	07 D	11 D	15 D	22 D	522 D/11			
E10	LevelControl Basic 2 control unit for single-pump station, IP54	BC1 230 DFNO 100	10,0	✗	-	-	-	-	-	-	19073760	4.5	
E11	LevelControl Basic 2 control unit for single-pump station DOL starting, with manual-0-automatic selector switch, indicator lamps and control panel, high water alert, integrated alarm buzzer 85 dB(A), operating hours counter/ start-stop cycles per pump, voltage measurement, phase monitoring, indication of water level, volt-free contact for general fault message, motor temperature warning (thermal circuit breaker) – self-acknowledging; motor leakage monitoring, optional mains-independent alarm via rechargeable battery (E90) For float switch or 4...20 mA sensor, optional master switch, 400 x 278 x 120 mm 400 V variant with motor protection switch 230 V variant with external socket	BC1 400 DFNO 025	2,5	-	✗	✗	-	-	-	-	19073762	4.5	
		BC1 400 DFNO 040	4,0	-	-	-	✗	✗	-	-	-	19073763	4.5
		BC1 400 DFNO 063	6,3	-	-	-	-	-	✗	✗	-	19073764	4.5
E90	Rechargeable battery retrofit kit for LevelControl Basic 2, type BC										19074194	0.8	
	Scope of supply: 2 rechargeable batteries (6 V, 1.3 Ah) and charge controller												
O1	Master switch for LevelControl Basic 2 BC, fitted, 3-pole, 20 A, lockable			✗	✗	✗	✗	✗	✗	✗	01143084	0.2	
O200	Signalling module for LevelControl Basic 2 BC Signalling module and IO extension module for LevelControl Basic 2 6 volt-free contacts: - Fault pump 1 - Fault pump 2 - High water - Sensor fault - Pump 1 in operation - Pump 2 in operation Analog output (fill level), Modbus interface for field bus module										19075182	0.2	

Control units for dual-pump units from the configurable range
Table 16: Control units for dual-pump units from the configurable range

Item	Description	Type	Current	AmaDrainer Box								Mat. No.	[kg]
				1~	3~								
				[A]	...E	05 D	07 D	11 D	15 D	22 D	522 D/11		
E30 ¹⁷⁾	LevelControl Basic 2 control unit for dual-pump station, IP54 Peak load operation, stand-by pump, DOL starting, with manual-0-automatic selector switch, indicator lamps and control panel, high water alert, integrated alarm buzzer 85 dB(A), operating hours counter/start-stop cycles per pump Voltage measurement, phase monitoring, volt-free contact for general fault message 400 V variant: with motor protection switch 230 V variant: with external socket Optional mains-independent alarm via re-chargeable battery with charge controller (E90) Optional master switch, 400 x 281 x 135 mm (2 float switches, 10 m, included in connection set)	BC2 230 DFNO 100	10,0	✗	-	-	-	-	-	-	19073774	4.7	
		BC2 400 DFNO 025	2,5	-	✗	✗	-	-	-	-	-	19073776	4.7
		BC2 400 DFNO 040	4,0	-	-	-	✗	✗	-	-	-	19073777	4.7
		BC2 400 DFNO 063	6,3	-	-	-	-	-	✗	✗	-	19073778	4.7
E90	Rechargeable battery retrofit kit for LevelControl Basic 2, type BC Scope of supply: 2 rechargeable batteries (6 V, 1.3 Ah) and charge controller										19074194	0.8	
O1	Master switch for LevelControl Basic 2 BC, fitted, 3-pole, 20 A, lockable			✗	✗	✗	✗	✗	✗	✗	01143084	0.2	
O200	Signalling module for LevelControl Basic 2 BC Signalling module and IO extension module for LevelControl Basic 2 6 volt-free contacts: - Fault pump 1 - Fault pump 2 - High water - Sensor fault - Pump 1 in operation - Pump 2 in operation Analog output (fill level), Modbus interface for field bus module										19075182	0.2	

¹⁷⁾ Scope of supply for the off-the-shelf range

Accessories

Unit accessories

Table 17: Overview of lifting unit accessories

Item	Description	Dimension	AmaDrainer Box				Mat. No.	[kg]
			1.1 U	1.1 B	Z2.1 U	Z2.1 B		
 P9	Pipe adapter, EPDM/stainless steel, PN 0,5 for connecting inlet lines and vent lines	DN 50 for outside pipe diameter 32 - 40 mm	-	-	X	X	01313527	0.11
		DN 50 for outside pipe diameter 53 - 63 mm	X	X	X	X	01313528	0.18
		DN 70 for outside pipe diameter 40 - 50 mm	X	X	X	X	01313529	0.36
		DN 70 for outside pipe diameter 53 - 63 mm	X	X	X	X	01313530	0.26
		DN 70 for outside pipe diameter 75 - 89 mm	X	X	X	X	01644617	0.28
 P11	Socket gate valve, CuZn PN 16 with internal thread and full port	Rp 1 1/2	X	X	-	-	00411502	0.8
		Rp 2	-	-	X	X	00411503	1.29
 P20	Plug for closing collecting tank connections that are not used in dual-pump lifting systems / operation with one pump set	Rp 2 1/2	-	-	X	X	05148425	0.2
 P33	Sealing collar with O-ring or adhesive flange for connecting to a site-supplied sealing sheet that provides a moisture barrier, (for example, beneath an insulation layer)	DF 400	X	-	-	-	18041235	1.4
 P33		DF 630	-	-	X	-	18041236	1.4
 P34	Puddle flange with worm drive hose clips, EPDM/stainless steel To protect the floor plate against penetrating groundwater, 3 m groundwater level max.	MK 400	X	-	-	-	01313465	1.5
		MK 630 with level-adjusting piece H = 300 mm	-	-	X	-	18041238	12.8
 P35	PE level-adjusting piece with O-ring	H = 300 mm	-	-	X	-	39023481	5.75
 P36	Venting device with activated carbon filter	DN 70 / DN 80 / DN 100	X	X	X	X	01315158	0.32
 D	Discharge line connection DN 40 incl. clamped connection	Rp 1 1/2 x 50 PE-HD 50	X	-	-	-	01052682	0.29
	Discharge line connection DN 50 incl. clamped connection	Rp 1 1/2 x 63 PE-HD 63	X	-	-	-	01155619	0.49
	Discharge line connection Dual-pump station DN 50 with clamped connection	Rp 2, PE-HD 63 PEHD 63	-	-	X	-	01052681	0.51

Alarm switchgears for pumps, non-ATEX-compliant

Table 18: AS 0 / AS 1 / AS 2 / AS 4 / AS 5 / AS W4 / AS W8

Item	Description	Mat. No.	[kg]
E50	 Alarm switchgear AS 0 With circuit breaker, acoustic signalling device with 85 dB(A), green equipment-on lamp Plastic housing, IP20, H x W x D = 140 x 80 x 57 [mm]. Use float switch, F1 leakage sensor (item E64), M1 alarm contactor or signal relay of control unit as contactor.	29128401	0.5
E51	 Alarm switchgear AS 2 With circuit breaker, acoustic signalling device with 85 dB(A), green equipment-on lamp, volt-free contact for hook-up to a control station Plastic housing, IP20, H x W x D = 140 x 80 x 57 [mm]. Use float switch, F1 leakage sensor (item E64) or signal relay of control unit as contactor.	29128422	0.5
E52	 Alarm switchgear AS 4 With circuit breaker, acoustic signalling device with 85 dB(A), green equipment-on lamp, volt-free contact for hook-up to a control station, self-charging power supply unit for 5 hours of operation in the event of a power failure Plastic housing, IP20, H x W x D = 140 x 80 x 57 [mm]. Use float switch (E60), F1 leakage sensor (item E64) or signal relay of control unit as contactor.	29128442	0.5
E53	 Alarm switchgear AS 5 Mains-independent, with self-charging power supply unit for 10 hours of operation in the event of a power failure, mains pilot LED, fault indicator light, acknowledgement button, volt-free contact for hook-up to a control station, ready for connection with 1.8 m power cable and plug. ISO housing, IP41, H x W x D = 190 x 165 x 75 [mm]. Use float switch (E60) or signal relay of control unit as contactor.	00530561	1.7
E55	 Alarm switchgear AS 1 In IP30 ISO plug housing, mains-independent, with self-charging power supply unit for 5 hours of operation in the event of a power failure, acoustic signalling device 70 dB(A) with circuit breaker and integrated signal transmitter with 3-metre power cable, max. 60 °C, not suitable for steam and condensate. 1. High water alert by suspending the sensor in a (pump) sump above the pump start-up point. 2. Water alert signal at a water level of only 1 mm by placing the sensor on the floor in areas with a flooding or leakage risk, e.g. the cellar or next to the washing machine in the kitchen or bathroom.	00533740	0.9
E330	 KSB washing machine interrupter Alarm triggered, with automatic interruption of power supply to the washing machine if the water level in the collecting tank is too high 230 V AC, 50/60 Hz, max. 16 A ohm resistive load Comprising: ▪ housing with socket and plug ▪ buzzer ▪ automatic re-start ▪ M1 alarm contactor with connection cable (5 m)	01318215	0.28
E331	 AS W4 / W8 alarm switchgear Alarm triggered with automatic interruption of power supply of up to 4 or 8 washing machines if the water level in the collecting tank is too high 230 V AC, 50/60 Hz, max. 16 A resistive load per intermediate connector Comprising: ▪ Control unit ▪ Buzzer ▪ Automatic re-start ▪ M1 alarm contactor with connection cable (5 m) ▪ 4 or 8 intermediate connectors		
	KSB AS W4 - KSB washing machine interrupter for up to 4 washing machines	19075394	4
	KSB AS W8 - KSB washing machine interrupter for up to 8 washing machines	19075395	4
E332	AS1-M alarm switchgear in ISO plug housing IP30, mains-independent, with self-charging power supply unit for 5 hours' operation in the event of a power failure, acoustic signal 70 dB(A), circuit breaker, sensor connection bush and integrated monitoring for broken wires in the sensor cable Max. ambient temperature: 60° C Complete with M1 insertion-type sensor and installation accessories for high-water alert 230 V~/9 V = 1,5 VA	19074516	0.5

Control unit / switchgear accessories
Table 19: Overview of control unit / switchgear accessories

Item	Description	Cable length	Mat. No.	[kg]
		[m]		
	Float switch with free cable end Function: circuit closed in upper float position (NO contact) Float switch housing: polypropylene Fluid temperature: ≤ 70 °C Power cable: H07RN-F3G1	3	11037742	0.5
		5	11037743	0.8
		10	11037744	1.3
	Float switch with free cable end, oil-resistant Function: circuit closed in upper float position (normally open contact) Float switch housing: polypropylene Fluid temperature: max. 70 °C Power cable: PUR 3×1.0 mm ²	5	11037753	0.8
		10	11037754	1.2
		20	11037755	2
	F1 leakage sensor As contactor for alarm switchgears AS 0, AS 2 or AS 4, with 3-metre power cable, max. 40 °C, not suitable for steam and condensate. Warning at a water level of 1 mm by placing the sensor on the floor in areas with a flooding or leakage risk, e.g. the cellar or next to the washing machine in the kitchen or bathroom.	3 m	19072366	0.2
	Horn, 12 V DC, 105 dB, 150 mA, IP54 Suitable for indoor installation and outdoor installation. Protect against moisture.		01086547	0.1
	Alarm combination (yellow lamp and piezo buzzer 92 dB), 12 V DC, 120 mA, IP65		01139930	0.1
	Safety switch STECKMAT trips in approx. 0.03 seconds, from approx. 0.03 A, 230 V / 10 A		00534217	0.5
	Rechargeable battery retrofit kit for LevelControl Basic 2, type BC Scope of supply: 2 rechargeable batteries (6 V, 1.3 Ah) and charge controller		19074194	0.8
	M 1 alarm contactor for AmaDrainer Box (100 l) Magnetic float switch as contactor for alarm switchgears AS 0, AS 2, AS 4 and AS 5 or LevelControl Basic 2 control unit	5	18040562	0.8



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